

Classification of Astronomical Objects (Stars, Galaxies, Quasars) using Artificial Neural Networks

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Abstract—This work studies supervised classification of astronomical objects into Stars, Galaxies and Quasars using three feed-forward Artificial Neural Network (ANN) variants: (i) a baseline ANN, (ii) a regularized ANN with dropout and L2 regularization, and (iii) an ANN with architecture optimized using a Genetic Algorithm. Classical machine learning models including Decision Trees and Random Forests were also implemented as benchmark classifiers for comparative evaluation. Feature engineering was applied uniformly across all models. The models were trained on a tabular photometric dataset consisting of 100,000 samples with 13 input features. Training was performed in Google Colab using TensorFlow/Keras with early stopping and model checkpointing. Performance was evaluated using accuracy, precision, recall, F1-score, confusion matrices, and generalization metrics. Results demonstrate that the regularized ANN achieved superior robustness and competitive accuracy compared to both classical machine learning and optimized neural models.

I. INTRODUCTION

Machine learning methods ranging from classical algorithms such as Decision Trees and Random Forests to deep learning approaches have been widely adopted for astronomical object classification. Tree-based ensemble models are particularly effective on structured tabular data due to their robustness to nonlinear feature interactions and noise. However, deep neural networks provide greater representational flexibility and allow the incorporation of regularization and explainability techniques to improve generalization.

This paper presents a detailed evaluation of three ANN variants alongside classical machine learning baselines. While Decision Tree and Random Forest classifiers serve as performance references, the primary focus is placed on optimizing and interpreting ANN-based models.

II. METHODOLOGY

A. Data Preprocessing

The preprocessing pipeline is based on the procedure described in the project report. The dataset consists of 100,000

photometric samples with features including u, g, r, i, z magnitudes, redshift proxy, and derived color indices such as $(u-g)$, $(g-r)$, $(r-i)$, and $(i-z)$. Preprocessing steps:

- Missing-value handling via median imputation.
 - Feature engineering: computing color indices and flux ratios as commonly used in astronomical catalogs.
 - Scaling using RobustScaler to handle long-tailed distributions and outliers.
 - Label encoding: classes encoded as {0: Star, 1: Galaxy, 2: Quasar}.
 - Stratified 60-20-20 split for train, validation, and test sets.
- All ANN variants described in this work were trained on the same feature-engineered dataset to ensure a fair comparison.

B. ANN Architecture

All ANN models operate on the same feature-engineered input consisting of 13 photometric attributes. The baseline and regularized ANNs share a common deep feed-forward structure, while the Genetic Algorithm (GA)-optimized ANN evolves its architecture automatically.

1) Baseline and Regularized ANN: The baseline ANN follows a fully connected deep architecture consisting of five hidden layers with progressively decreasing neuron counts:

- Input Layer: 13 neurons
- Hidden Layer 1: Dense(256, ReLU)
- Hidden Layer 2: Dense(128, ReLU)
- Hidden Layer 3: Dense(64, ReLU)
- Hidden Layer 4: Dense(32, ReLU)
- Hidden Layer 5: Dense(16, ReLU)
- Output Layer: Dense(3, softmax)

For the regularized ANN, batch normalization was applied after each hidden layer, along with dropout rates ranging from 0.2 to 0.3. L2 kernel regularization with a coefficient of 1×10^{-4} was incorporated to mitigate overfitting.

2) *GA-Optimized ANN*: The GA-optimized ANN evolved a three-hidden-layer architecture comprising 64, 32, and 256 neurons respectively. Each hidden layer employed ReLU activation followed by batch normalization, dropout with a rate of 0.115, and L2 regularization of 3.86×10^{-5} . The output layer consisted of three neurons with softmax activation.

C. Training Pipeline

Training was performed on Google Colab using TensorFlow/Keras:

- Loss: Categorical Cross-Entropy
- Optimizer: Adam (lr = 0.001)
- Epochs: up to 100 with EarlyStopping (patience 10)
- Batch size: 64
- Callbacks: EarlyStopping, ModelCheckpoint (saving best model based on validation accuracy)

D. Optimization and Explainability Techniques

To further enhance performance and interpretability, the following techniques were incorporated:

- **Particle Swarm Optimization (PSO)**: Used for hyperparameter tuning including learning rate, neuron counts, and dropout values.
- **Genetic Algorithm (GA)**: Applied to search optimal ANN architectures (number of layers and neurons).
- **SHAP (SHapley Additive exPlanations)**: Used to analyze global and local feature importance.
- **LIME (Local Interpretable Model-agnostic Explanations)**: Used for instance-level prediction explanations.

III. RESULTS

This section presents dataset characteristics, experimental environment, and the performance of all ANN variants.

A. Dataset

The experiments were conducted using the Stellar Classification Dataset derived from the Seventeenth Data Release (DR17) of the **Sloan Digital Sky Survey (SDSS)** [1]. The dataset contains approximately **100,000** astronomical observations classified as stars, galaxies, or quasars.

Each observation includes photometric magnitudes in the u, g, r, i, z bands, redshift measurements, and additional positional and observational metadata such as right ascension and declination. The original dataset comprises 17 feature attributes and one class label. For this study, 13 relevant photometric and spectral features were selected and further processed through feature engineering techniques (including color indices) prior to model training.

Stratified 60–20–20 splitting was applied to preserve class distributions across training, validation, and test sets.

B. Experiment Environment

All experiments were conducted in Google Colab using TensorFlow/Keras with EarlyStopping and ModelCheckpoint.

C. Decision Tree Classifier

A Decision Tree model was trained on the feature-engineered dataset to provide a classical machine learning baseline.

TABLE I: Decision Tree Classification Performance (Test Set)

Class	Precision	Recall	F1-score
Star	0.97	0.98	0.98
Galaxy	0.95	0.90	0.92
Quasar	1.00	0.99	1.00
Accuracy	0.9704		

The trained Decision Tree achieved a depth of 15 with 514 leaf nodes, indicating a highly branched structure. While strong performance was obtained, the model exhibited sensitivity to class imbalance, particularly for galaxy classification.

D. Random Forest Classifier

A Random Forest ensemble model was implemented to improve robustness over the single Decision Tree through bagging and ensemble learning.

TABLE II: Random Forest Classification Performance (Test Set)

Class	Precision	Recall	F1-score
Star	0.98	0.99	0.98
Galaxy	0.97	0.93	0.95
Quasar	0.99	1.00	0.99
Accuracy	0.9786		

The Random Forest achieved higher accuracy than the Decision Tree due to ensemble averaging. An overfitting gap of 0.0107 was observed between training and test performance, indicating mild overfitting despite improved stability. The model also achieved low test log loss (0.0777), reflecting confident probabilistic predictions.

E. Artificial Neural Networks (ANN)

Three ANN variants were evaluated using the same feature-engineered dataset.

1) *Baseline ANN*: The baseline ANN is a simple feed-forward neural network designed to establish a performance reference without explicit regularization techniques. It consists of three hidden layers with progressively decreasing neuron counts.

- Input Layer: 13 neurons (feature-engineered photometric attributes)
- Hidden Layer 1: Dense(64, ReLU)
- Hidden Layer 2: Dense(32, ReLU)
- Hidden Layer 3: Dense(16, ReLU)
- Output Layer: Dense(3, softmax)

The model was trained using the Adam optimizer and categorical cross-entropy loss for multi-class classification.

TABLE III: Baseline ANN Classification Performance

Class	Precision	Recall	F1-score
Star	0.9799	0.9765	0.9782
Galaxy	0.9603	0.9414	0.9508
Quasar	0.9706	0.9965	0.9834
Accuracy	0.9742		

2) *ANN with Dropout and Regularization*: This variant incorporates dropout, batch normalization, and L2 regularization to improve generalization performance.

The regularized ANN employs a deep feed-forward architecture consisting of five hidden layers with progressively decreasing neuron counts:

- Input Layer: 13 neurons
- Hidden Layers: 256, 128, 64, 32, and 16 neurons with ReLU activation
- Output Layer: 3 neurons with softmax activation

Dropout was applied with rates of 0.3 for the first two hidden layers and 0.2 for the subsequent layers, while L2 regularization with a coefficient of 1×10^{-4} was used to mitigate overfitting. Batch normalization was applied after each hidden layer. The model was trained using the Adam optimizer with a learning rate of 0.001 and batch size of 64.

The network contains 48,355 trainable parameters and 49,315 total parameters across 14 layers.

TABLE IV: Regularized ANN Classification Performance

Class	Precision	Recall	F1-score	Support
Star (0)	0.9783	0.9808	0.9795	10052
Galaxy (1)	0.9688	0.9322	0.9501	3261
Quasar (2)	0.9745	1.0000	0.9871	3714
Accuracy	0.9757			

The regularized ANN exhibited excellent generalization behavior with an overfitting gap of only 1.5×10^{-4} between training and validation accuracy.

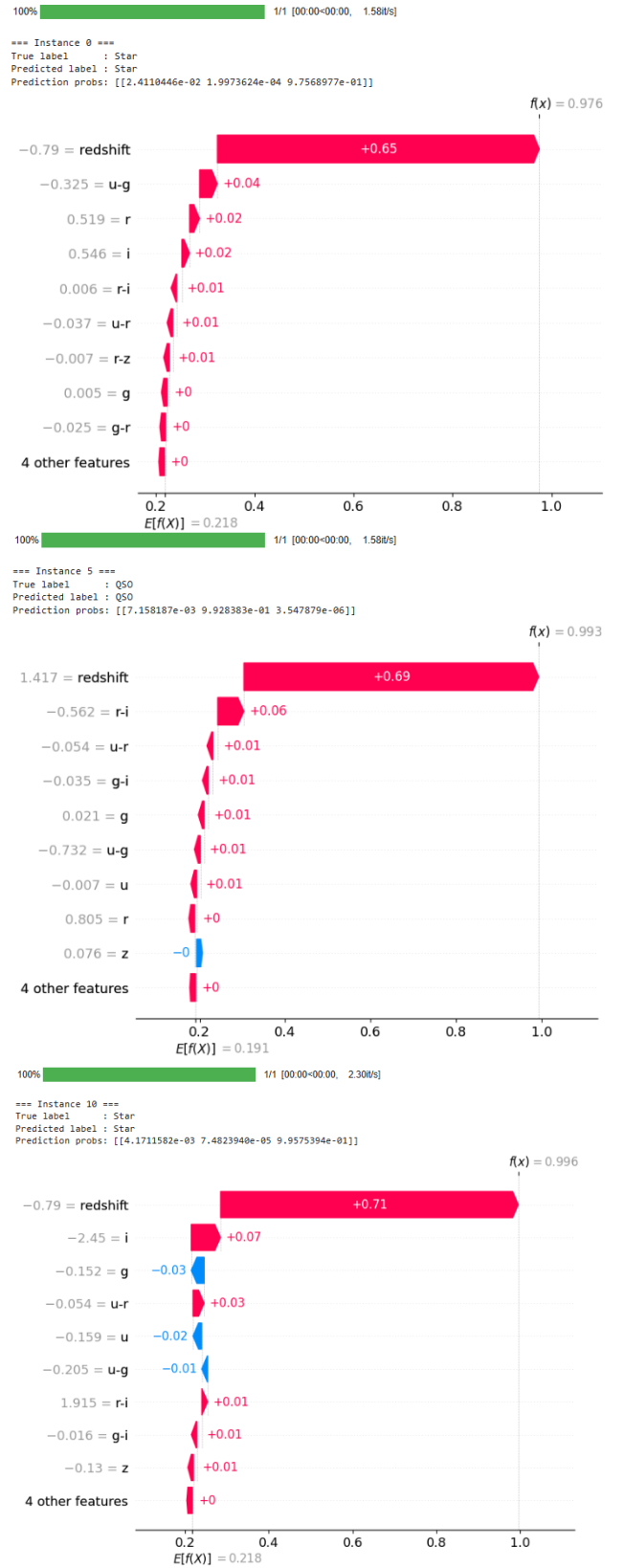


Fig. 1: SHAP feature importance plots for the Regularized ANN

a) SHAP Interpretation — Regularized ANN:

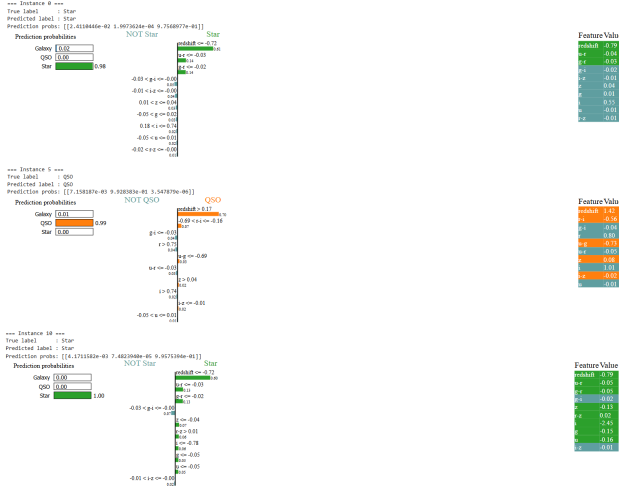


Fig. 2: LIME explanations for individual predictions using the Regularized ANN

b) LIME Interpretation — Regularized ANN:

3) ANN Optimized using Genetic Algorithm: A Genetic Algorithm (GA) was employed to automatically evolve the ANN architecture by optimizing the number of hidden layers and neurons to balance performance and model complexity.

Compared to the regularized ANN, the GA-optimized model achieved a significantly more compact structure, reducing the number of dense layers from six to three and the total trainable parameters from 48,355 to 12,483, while maintaining comparable classification accuracy.

TABLE V: GA-Optimized ANN Classification Performance

Class	Precision	Recall	F1-score
Star	0.9785	0.9778	0.9782
Galaxy	0.9679	0.9341	0.9507
Quasar	0.9677	0.9992	0.9832
Accuracy	0.9741		

The GA-optimized ANN achieved a training accuracy of 0.9727, validation accuracy of 0.9716, and test accuracy of 0.9704, with a minimal overfitting gap of 0.0011, indicating strong generalization performance.

TABLE VI: ANN Architecture Complexity Comparison

Model	Total Layers	Dense Layers	Trainable Params	Non-Trainable Params	Total Params
Regularized ANN	14	6	48,355	960	49,315
GA-Optimized ANN	7	3	12,483	576	13,059

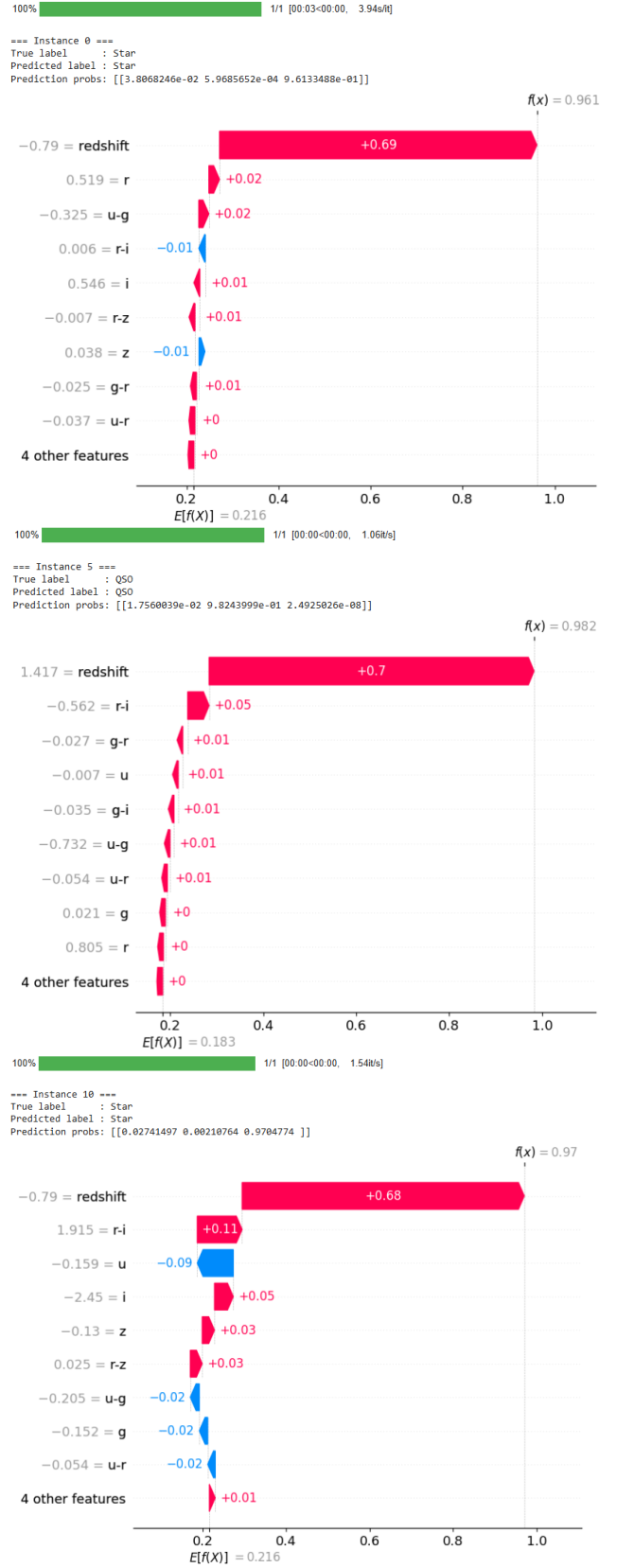


Fig. 3: SHAP feature importance plots for the GA-optimized ANN

a) SHAP Interpretation — GA-Optimized ANN:



Fig. 4: LIME explanations for individual predictions using the GA-optimized ANN

b) *LIME Interpretation — GA-Optimized ANN:* The GA converged to a compact three-layer architecture while maintaining high classification accuracy.

TABLE VII: Comparative Performance of ANN Variants

Model	Accuracy	Macro F1	Weighted F1	Key Characteristics
Baseline ANN	0.9742	0.9708	0.9741	No explicit regularization
Regularized ANN	0.9757	0.9723	0.9756	Dropout + L2 + BN
GA-Optimized ANN	0.9741	0.9707	0.9740	GA-optimized architecture

4) *Comparison of Classical Machine Learning Models with ANN-Based Approaches:* To evaluate the effectiveness of deep learning models over traditional machine learning techniques, the performance of Decision Tree and Random Forest classifiers was compared with the ANN variants using test accuracy as the primary metric.

TABLE VIII: Comparison of Classical ML Models and ANN Variants

Model	Test Accuracy
Decision Tree	0.9704
Random Forest	0.9786
Baseline ANN	0.9742
Regularized ANN	0.9757
GA-Optimized ANN	0.9741

5) *Justification for Selecting Regularized ANN over Random Forest:* Although the Random Forest classifier achieved the highest test accuracy (0.9786) among all evaluated models, further analysis revealed that the Regularized ANN provided superior generalization stability and robustness.

The Random Forest exhibited a noticeable overfitting gap of 0.0107 between training and test accuracy, indicating partial memorization of training data despite ensemble averaging. In contrast, the Regularized ANN demonstrated an extremely small overfitting gap of only 1.5×10^{-4} , reflecting highly consistent performance across training, validation, and test sets.

Moreover, the Random Forest learning curve showed persistently higher training accuracy relative to cross-validation accuracy, confirming mild overfitting behavior even as training data increased. The corresponding log loss values (0.0777 on the test set) further suggested reduced confidence calibration compared to the ANN models.

From a modeling perspective, the Random Forest consisted of 200 trees with an average of over 1,800 nodes per tree, resulting in more than 360,000 total nodes, significantly increasing structural complexity. In contrast, the Regularized ANN achieved competitive accuracy using a controlled deep architecture with only 49,315 total parameters, benefiting from dropout, batch normalization, and L2 regularization to suppress overfitting.

An additional advantage of the ANN framework was the ability to apply post-hoc interpretability techniques such as SHAP and LIME, which provided both global and local explanations of feature influence. These tools revealed that photometric color indices and redshift-related features dominated classification decisions, aligning with astrophysical expectations. Such explainability is not inherently available in traditional Random Forest models.

Overall, while Random Forest achieved marginally higher raw accuracy, the Regularized ANN offered:

- Superior generalization stability (minimal overfitting gap)
- Controlled model complexity
- Strong predictive performance
- Integrated explainability through SHAP and LIME

For these reasons, the Regularized ANN was selected as the final preferred model for astronomical object classification.

6) *Comparative Discussion:*

- All ANN variants achieved classification accuracy above 97%.
- The regularized ANN demonstrated the best generalization performance.
- GA-based optimization produced a compact architecture with competitive accuracy.

- SHAP and LIME analyses revealed that photometric color indices and magnitude differences were the most influential features.

7) *Selection of Final Model: Regularized ANN vs Genetic Algorithm ANN*: To identify the most reliable ANN variant for deployment, a quantitative comparison was performed between the regularized ANN and the GA-optimized ANN using three criteria: test accuracy, test loss, and generalization gap (difference between training and validation accuracy).

The selection logic prioritized higher classification accuracy, followed by lower test loss, and finally lower overfitting gap to ensure robust generalization. This evaluation consistently favored the regularized ANN as the superior model.

The regularized ANN achieved the highest test accuracy while maintaining low loss and stable performance across training and validation phases. Although the GA-optimized ANN produced a more compact architecture, its predictive performance was marginally lower.

Based on this multi-criteria evaluation, the regularized ANN was selected as the final preferred model for astronomical object classification.

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